Technical Report

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Summary

This report analyses NHS data to evaluate resource adequacy and utilisation, using data wrangling, visualisation, and trend identification. Key findings inform recommendations for improving staffing, scheduling, and service delivery.

Introduction

Background

The NHS, the UK's publicly funded healthcare system, must expand its infrastructure to meet a growing population. Stakeholders debate whether current capacity is sufficient. The data analyst team is tasked with exploring available data and providing recommendations based on their findings.

Business Question

- Has there been adequate staff and capacity in the networks?
- What was the actual utilisation of resources?

Objectives

- To assess the adequacy of staffing and resources in the NHS networks.
- To determine the actual utilisation of resources in the NHS.
- To identify trends and patterns in the utilisation of NHS resources.
- To analyse the potential value of using external data sources, such as Twitter, for decision-making.
- To provide data-driven recommendations to the NHS based on analysis.

Analytical Approach

Data Collection

At the project's start, I received three files: actual_duration.csv, appointments_regional.csv, and national_categories.xlsx. I also noted missing data between June-November 2018. Additionally, I was provided with pre-scraped Twitter (now X) data in tweets.csv.

Data Analysis

Data Wrangling

- I loaded the CSV and Excel files into Python using the relevant Pandas functions, and made a DataFrame for each.

- The data was semi-wrangled, but I used *isna ()* and *dropna()* to identify and handle missing data.
- Additionally, I adjusted data types of columns (e.g. 'appointment_date ') for consistency and analytical preparation, using functions such as *astype()*, *to_datetime*.
- I also used Pandas data aggregation by using *groupby()*. Created more DataFrames from the original datasets, to explore trends further.
- I also filtered datasets to extract subsets, based on criteria for focused analysis.

Exploratory Data Analysis (EDA)

- For EDA I used visualisations, such as:
    - line plots for trends over time.
    - bar plots to compare categories.
    - box plots to examine distributions.
- Analysing trends in appointment numbers, missed appointments and service utilisation to understand seasonal trends.
- I compared different categories of appointment types, to find relative performance and significance of each, mainly looking at the appointment numbers to compare:
    - Service settings.
    - Context types.
    - Healthcare professional types ('hcp_type')

Descriptive Statistics

- Using the *count()* function I regularly counted the number of appointments, locations and more
- Using mean and standard deviation to identify the range of data, to understand limitations.
    - Primarily used in the *describe()* function
- To identify the range of the data sets dates, I used *min()* and *max()* functions.

Visualisation and Insights

Visualisations

Figure 1

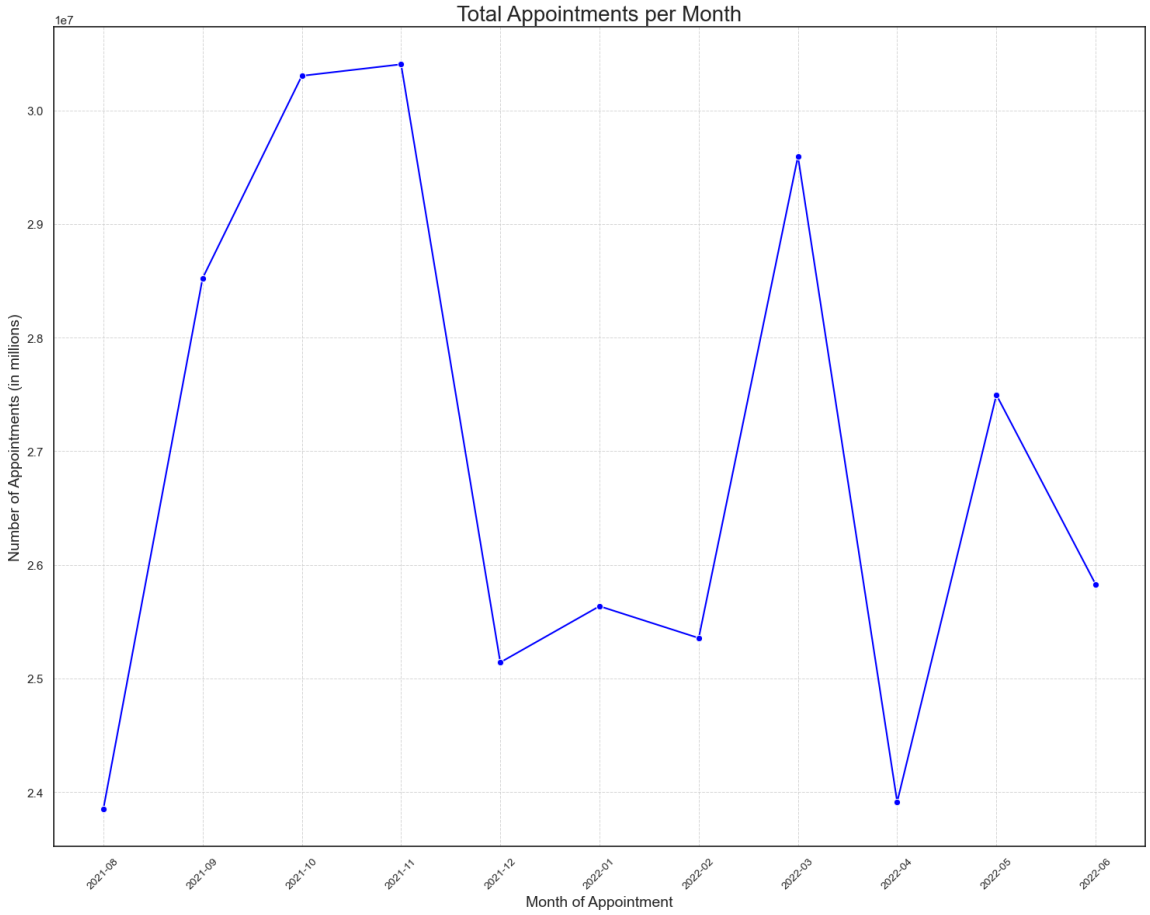


Figure 2

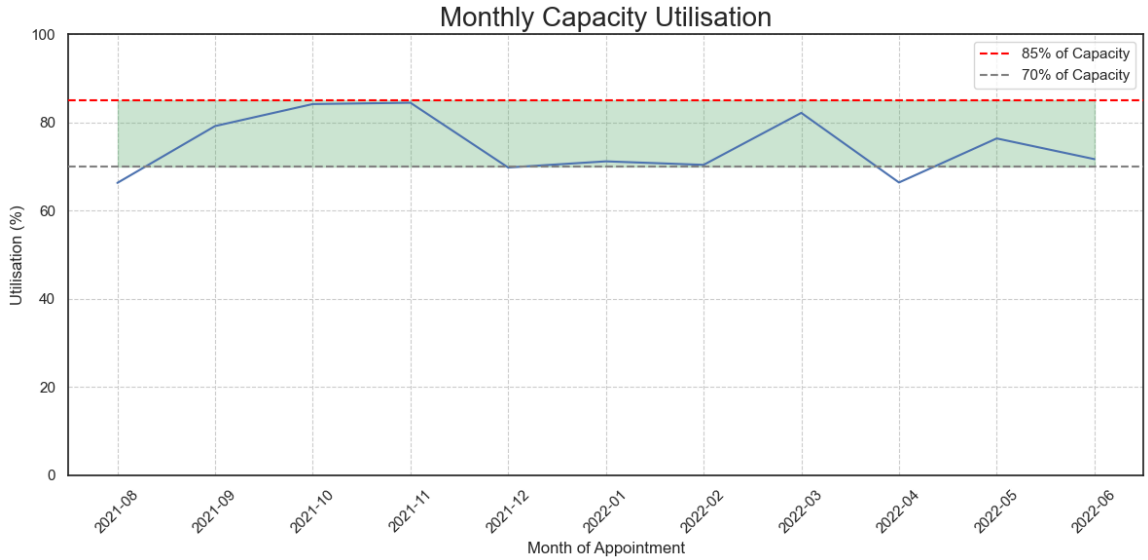


Figure 3

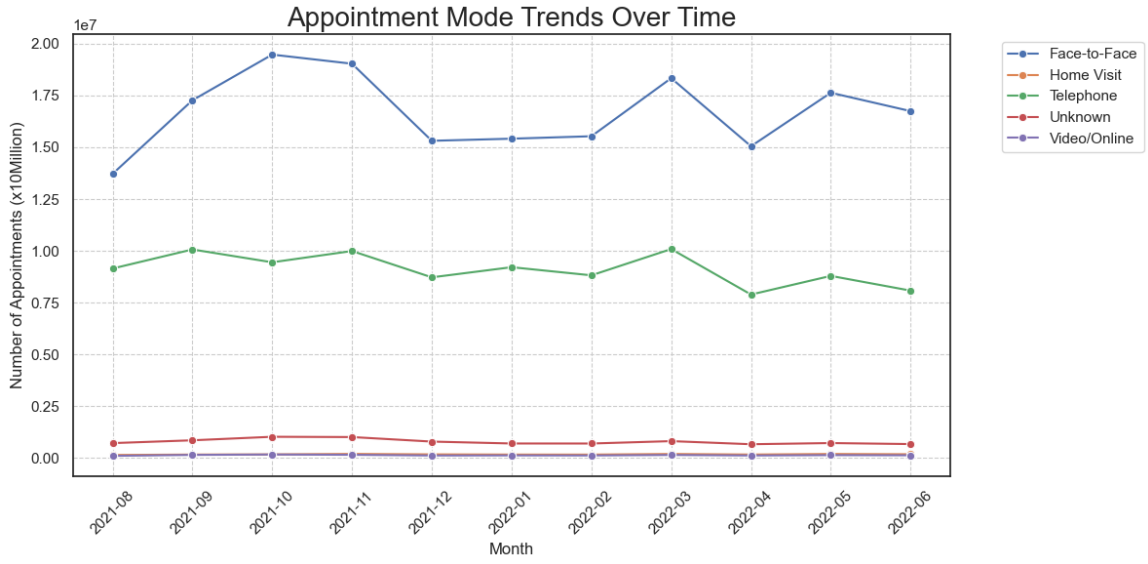


Figure 4

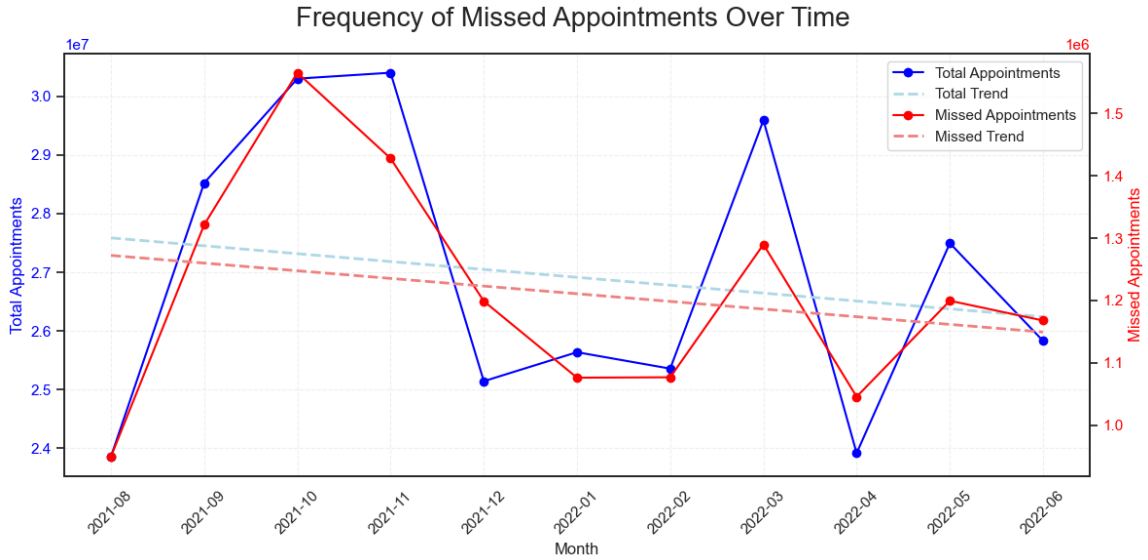


Figure 5

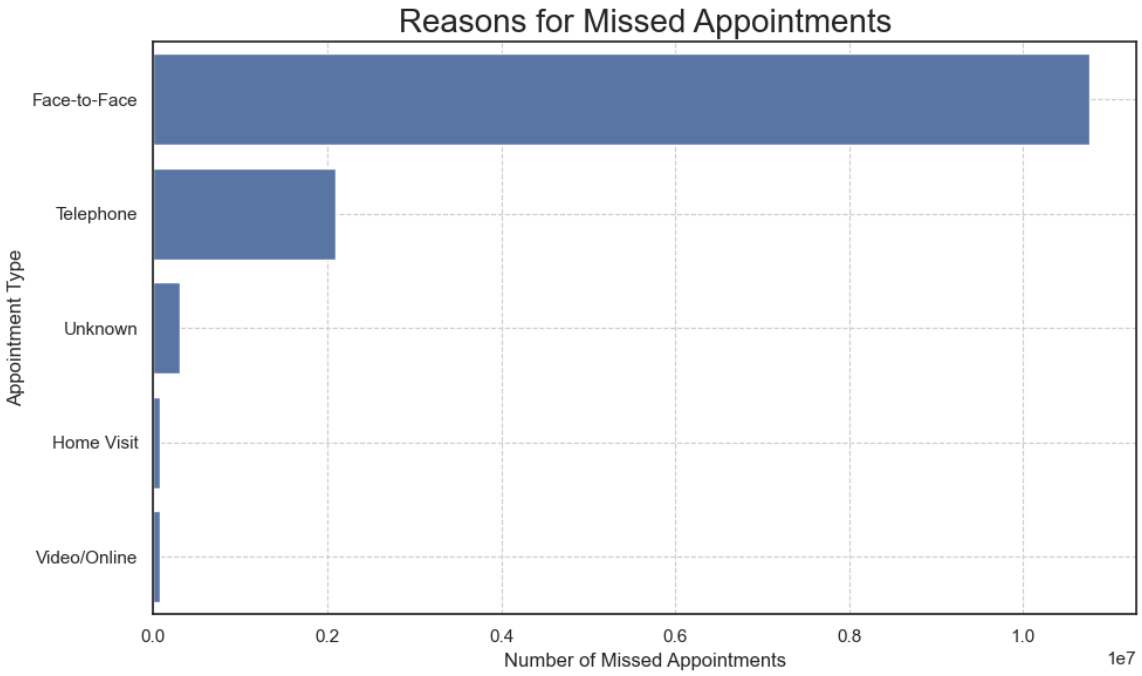


Figure 6

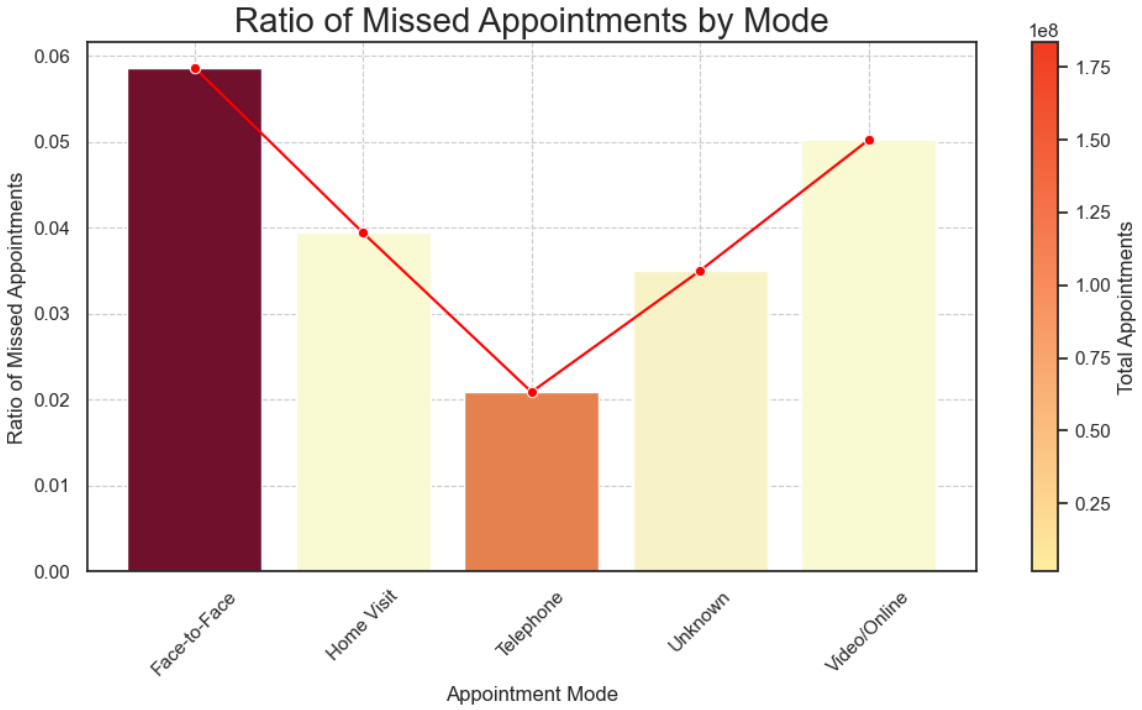


Figure 7

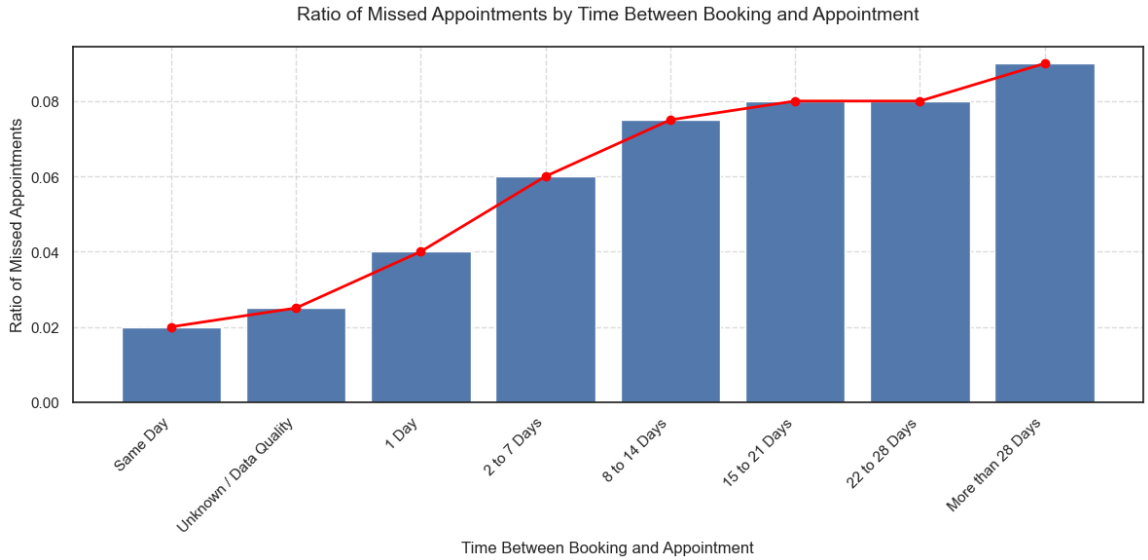


Figure 8

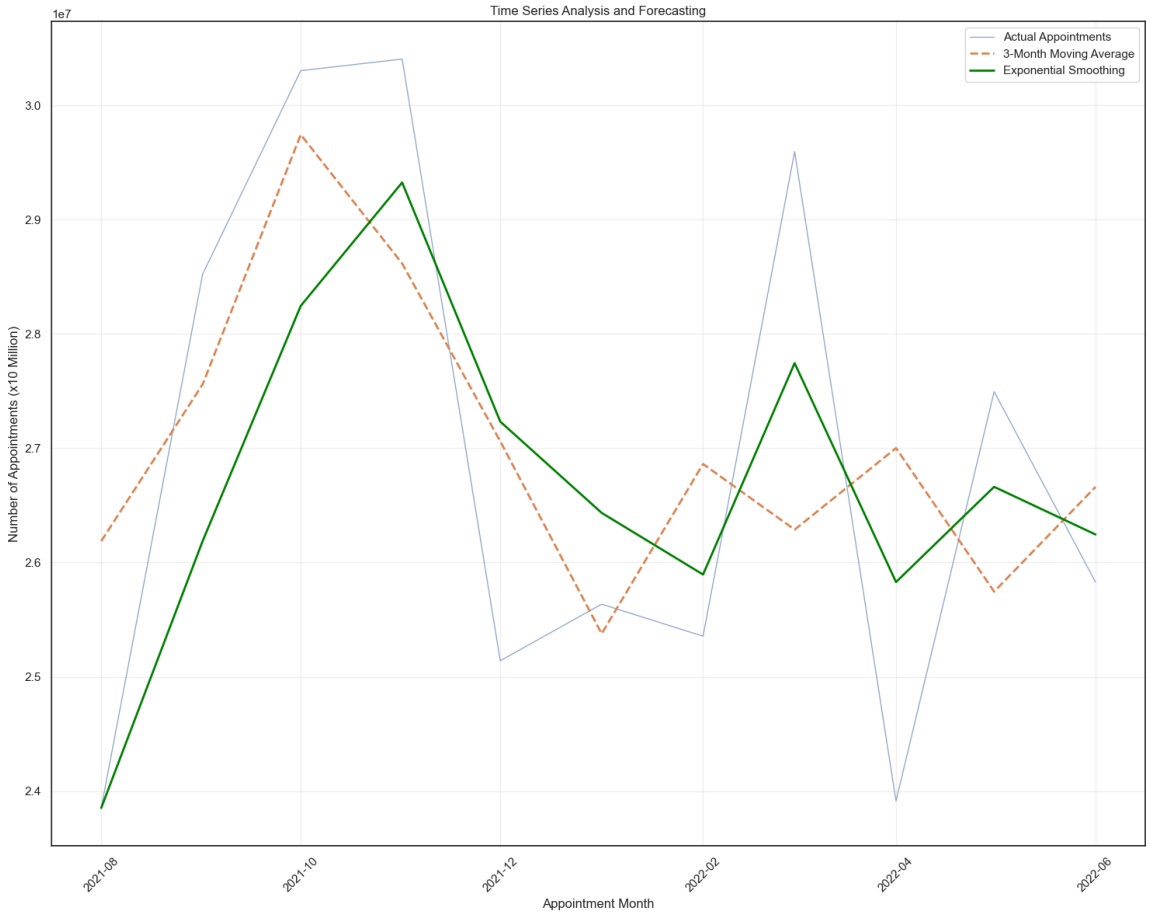
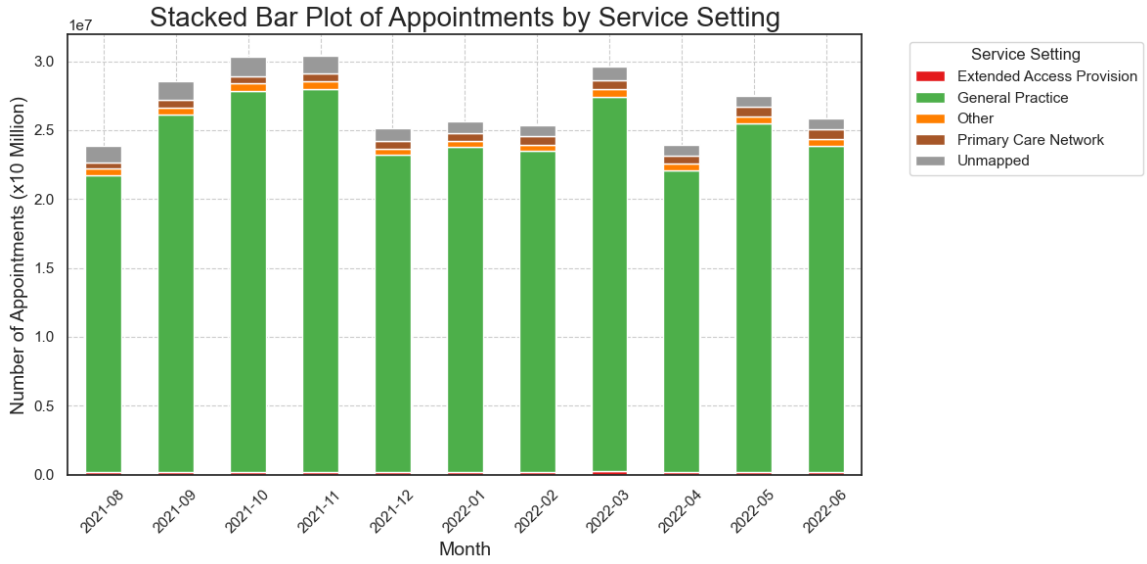


Figure 9



Interpretation of Visualisations

- **Total Appointments per Month** (figure 1)**:** The line plot shows a fluctuation in total appointments over the months, with a noticeable dip in December 2021 and January 2022. This dip could coincide with holiday seasons or potentially be attributed to factors like increased awareness of alternative appointment modes.
- **Monthly Capacity Utilisation** (figure 2)**:** This visualisation tracks the NHS's capacity utilisation over time. A red dashed line at 85% indicates a critical threshold. The plot shows that utilisation frequently approaches or exceeds this 85% capacity mark, especially from September to November 2021, highlighting periods of potential strain on resources.
- **Appointment Mode Trends Over Time** (figure 3)**:** This plot illustrates the dominance of face-to-face appointments while demonstrating the consistent, though lower, utilisation of telephone and video/online consultations.
- **Frequency of Missed Appointments Over Time** (figure 4)**:** The line plot shows a fluctuating trend of missed appointments, peaking in October 2021 and showing a decline thereafter. This suggests potential success of interventions aimed at reducing missed appointments.
- **Reasons for Missed Appointments** (Figure 5)**:** This bar plot reveals that face-to-face appointments have the highest number of missed appointments. However, it's important to note that this doesn't consider the proportion of missed appointments relative to the total number of appointments in each mode.
- **Ratio of Missed Appointments by Mode** (Figure 6)**:** While the previous bar plot showed face-to-face as having the most missed appointments, this visualisation, by considering the ratio, reveals that telephone consultations have the lowest proportion of missed appointments, followed by video/online.
- **Ratio of Missed Appointments by Time Between Booking** (Figure 7)**:** This plot indicates that appointments booked on the same day ("1 Day") have the lowest

missed appointment ratio. The ratio generally increases as the time between booking and appointment lengthens.

- **Time Series Analysis and Forecasting** (Figure 8)**:** This plot presents a time series analysis of appointment counts, employing moving averages and exponential smoothing techniques to visualise trends and potential future projections.
- **Stacked Bar Plot of Appointments by Service Setting** (Figure 9)**:** This plot offers a comprehensive view of how different service settings contribute to the overall volume of appointments. The stacking visually emphasises the proportion of each service setting in the total monthly appointments.

Patterns and Trends

Identified Patterns Implications

- Capacity and Staffing: The "Monthly Capacity Utilisation" plot shows frequent utilisation above 85%, highlighting resource strain and suggesting increased staffing during peak months (e.g., September to November). Additionally, high volumes of same-day appointments suggest a need for improved demand management, possibly through optimised scheduling or incentives for early bookings.
- Modes and Missed Appointments: Face-to-face consultations remain the preferred mode, but they have the highest number of missed appointments. Telephone and video/online options show lower missed appointment rates, suggesting targeted interventions, like reminders, could further reduce missed rates for these modes.
- Service Setting Impact: General Practice dominates appointment volumes, but excluding it shows the critical role of settings like "Extended Access Provision" and "Primary Care Network" in handling peak demand, stressing the need for their support. Additionally, addressing "Unmapped" data is essential for improving overall data quality and understanding service utilisation.
- Seasonal Trends: Appointment trends show predictable seasonal peaks, allowing proactive resource planning, especially for high-demand winter months. Time series forecasting further supports long-term planning by projecting future demand.

Recommendations

- Optimise Staffing & Capacity:
    - Increase staff levels in high-utilisation areas and review the current skill mix to ensure appropriate staffing ratios.
    - Implement flexible staffing models to adjust for real-time demand, such as using on-call staff during peak periods.
- Improve Scheduling & Reduce Missed Appointments:

    - Develop a robust demand management system to optimise appointment slots and reduce same-day pressures.
    - Implement targeted appointment reminders and incentivise advance bookings to decrease missed appointments.
- Enhance Alternative Appointment Modes:
    - Promote telephone and video consultations to improve access and reduce wait times.
    - Improve the virtual consultation experience with user-friendly platforms and technical support for patients.

Conclusion

The NHS experiences capacity pressures, particularly during peak seasons, and patients generally prefer face-to-face consultations. Telephone and video options have lower missed rates. Extended Access Provision services help manage demand. Improved data quality, proactive staffing, optimised scheduling, and promoting virtual modes are key recommendations for better service delivery.

Appendices

- Example of importing using Pandas:

Prepare your workstation

[320]:
```
# Import the necessary libraries.
import pandas as pd
import numpy as np
from datetime import datetime

# Optional - Ignore warnings.
import warnings
warnings.filterwarnings('ignore')
```

Loading and cleaning the 3 datasets:

- 'actual_duration.csv',
- 'actual_duration_2.csv',
- 'actual_duration_3.csv'

- Example of isna:

[224]:
```
# Determine whether there are missing values.
ad_na = ad[ad.isna().any(axis=1)]

# Are ther any missing values.
print("The number of missing values in the 'national_categories.xlsx' file:")
print(len(ad_na))
```

```
The number of missing values in the 'national_categories.xlsx' file:
0
```

- Example of astype():

[109]:
```
# Convert the appointment_month.
ar_df['appointment_month'] = ar_df['appointment_month'].astype(str)


# Create a lineplot with Seaborn.
sns.lineplot(data=ar_df, x='appointment_month', y='count_of_appointments', markers=True, marker='o', color='blue')

# Set the title and labels
plt.title('Total Appointments per Month', fontsize=20)
plt.xlabel('Month of Appointment', fontsize=14)
plt.ylabel('Number of Appointments (in millions)', fontsize=14)

# Rotate x-axis labels.
plt.xticks(rotation=45, fontsize=10)

# Add a grid to the plot with customized style
plt.grid(True, linestyle='--', linewidth=0.7, alpha=0.8)

# Display the plot.
plt.tight_layout()
plt.show()
```

- Example of groupby():

```
[ ]: # Create a DataFrame.
time_trends = ar_agg.groupby(['appointment_month', 'time_between_book_and_appointment'])['count_of_appointments'].sum().reset_index()

# View the DataFrame.
time_trends.head()
```

```
[ ]: # Set the plot size.
plt.figure(figsize=(12, 6))

# Create a line plot.
sns.lineplot(
    data=time_trends,
    x='appointment_month',
    y='count_of_appointments',
    hue='time_between_book_and_appointment',
    marker='o'
)

# Add titles and labels.
plt.title('Number of Appointments by Time Between Booking and Appointment Over Time')
plt.xlabel('Month')
plt.ylabel('Number of Appointments')
plt.xticks(rotation=45)
plt.xticks(rotation=45)

# Add a grid to the plot.
plt.grid(True, linestyle='--', linewidth=0.8)

# Move legend outside
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

# Show the plot.
plt.tight_layout()
plt.show()
```

- Example of filtering:

```
[ ]: # Group data to get the total number of appointments for each time_between_book_and_appointment.
total_appointments_by_hcp = ar_agg.groupby('time_between_book_and_appointment')['count_of_appointments'].sum().reset_index()
total_appointments_by_hcp.columns = ['time_between_book_and_appointment', 'total_appointments']

# Filter for missed appointments and group by hcp_type.
missed_appointments_by_hcp = missed_appointments.groupby('time_between_book_and_appointment')['count_of_appointments'].sum().reset_index()
missed_appointments_by_hcp.columns = ['time_between_book_and_appointment', 'missed_appointments']

# Merge the total and missed appointment counts.
wait_time = pd.merge(missed_appointments_by_hcp, total_appointments_by_hcp, on='time_between_book_and_appointment')

# Calculate the ratio of missed appointments.
wait_time['missed_ratio'] = wait_time['missed_appointments'] / wait_time['total_appointments']

wait_time.head()
```